

Classical Physics

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Part I

Classical mechanics

Chapter 1

Analytical mechanics

1.1 Lagrangian mechanics

Newtonian mechanics

1.1 (Laws of motion). Galilean structure, Galilean group

1.2 (Conservation laws).

Calculus of variations

1.3 (Euler-Lagrange equation).

1.4 (Closed system). $\frac{\partial \mathcal{L}}{\partial t} = 0$

1.5 (Definition of generalized momentum). $\frac{\partial \mathcal{L}}{\partial q} = 0$

1.6 (Equivalence to Newtonian mechanics).

Rigid bodies

1.7 (Inertia tensor).

1.8 (Eulerian angle).

1.9 (Lagrangian top).

Oscillation

1.10 (Harmonic oscillator).

1.11 (Damped oscillation).

1.12 (Pendulum).

1.13 (Lissajous curve).

1.14 (Coupled oscillation).

Central forces

1.15 (Polar coordinates).

1.16 (Effective potential).

1.17 (Kepler's problem).

1.18 (Rutherford scattering).

System of particles

1.19 (Closed systems).

1.20 (Collisions).

1.21 (Two-body problem).

1.22 (Three-body problem).

Euler-Lagrange equations

1.23 (Brachistochrone).

1.24 (Geodesic on the sphere).

1.25 (Dido's isoperimetric problem).

1.26 (Pendulum with moving support). A rheonomic system

1.27 (Sliding beads on a rim).

1.28 (Double pulley system).

1.2 Hamiltonian mechanics

Chapter 2

Continuum mechanics

2.1 Conservation laws

2.2 Fluid mechanics

2.3 Solid mechanics

plasticity, elasticity?

Chapter 3

Statistical mechanics

3.1 Thermodynamics

Laws of thermodynamics Equation of states Maxwell's relations
Thermal processes

3.2 Kinetic theory

ergodic hypothesis Boltzmann statistics Boltzmann equation, chapman enskog BBGKY hierarchy stochastic processes linear response

3.3 Ensembles

ensembles microcanonical, canonical, grand canonical classical gas Boltzmann distribution
x Two statistics x Fermi sea x Bose-Einstein condensation

Part II

Classical field theory

Chapter 4

Relativity

4.1 Special relativity

4.2 General relativity

4.3 Einstein field equation

4.4 Black holes

Chapter 5

Electromagnetism

5.1 Maxwell equations

We use the east coast sign convention and the Einstein summation convention. Let $M := \mathbb{R}^{1,3}$ be the Minkowski space. Consider a line bundle L over M and take an open subset U of M on which the bundle is trivialized.

A section of L describes...? Why is the external current J in $\Omega^3(U, \mathfrak{g})$?

A connection of L describes a photon field.

The Maxwell equation is the equation of motion of electromagnetic potential A and electromagnetic field F , and the inhomogeneous version is written as

$$d * F = \mu_0 J \quad : \quad \partial_\nu F^{\mu\nu} = \mu_0 J^\mu,$$

where $F \in \Omega^2(U, \mathfrak{g})$ and $J \in \Omega^3(U, \mathfrak{g})$ such that

$$F := dA + A \wedge A \quad : \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

where $A \in \Omega^1(U, \mathfrak{g})$. Note that we always have $A \wedge A = 0$ because $\mathfrak{g}$ is abelian. Then,

$$\begin{aligned} F &= dA = d(A_\mu dx^\mu) \\ &= dA_\mu \wedge dx^\mu + A_\mu d^2 x^\mu \\ &= \partial_\nu A_\mu dx^\nu \wedge dx^\mu \\ &= \partial_\nu A_\mu (dx^\nu \otimes dx^\mu - dx^\mu \otimes dx^\nu) \\ &= (\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \otimes dx^\nu. \end{aligned}$$

or

$$\begin{aligned} F(X, Y) &= \partial_\nu A_\mu (dx^\nu \wedge dx^\mu)(X, Y) \\ &= \partial_\nu A_\mu (X^\nu Y^\mu - Y^\nu X^\mu) \\ &= \partial_\nu A_\mu X^\nu Y^\mu - \partial_\nu A_\mu Y^\nu X^\mu \\ &= \partial_\mu A_\nu X^\mu Y^\nu - \partial_\nu A_\mu X^\mu Y^\nu \\ &= (\partial_\mu A_\nu - \partial_\nu A_\mu) X^\mu Y^\nu. \end{aligned}$$

The Maxwell equation is given by the Yang-Mills action

$$S[A] = \int_M \text{tr} \left(-\frac{1}{\mu_0} F \wedge *F - A \wedge J \right) \quad : \quad S = \int d^4x \left[-\frac{1}{\mu_0} F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu \right],$$

where J is given as an external current.

The anti-symmetry of F implies that $\partial_\mu \partial_\nu F^{\mu\nu} = 0$, so we have the charge conservation $\partial_\mu J^\mu = 0$.
quantum fluctuation of continuous field is interpreted as a particle

- Poincaré symmetry
- Gauge symmetry
- locality

$$A_\mu \mapsto A'_\mu = A_\mu + \partial_\mu \alpha$$

For a local gauge transform $g \in \Omega^0(U, G)$, by taking the logarithm suitably, we can identify g to $\alpha \in \Omega^0(U, \mathfrak{g})$ which satisfies $\exp \alpha = g$, and we have $d\alpha \in \Omega^1(U, \mathfrak{g})$.

Since $A \in \Omega^1(U, \mathfrak{g})$ is in fact a connection form $\omega \in \Omega^1(P|_U, \mathfrak{g})$ such that ... , the gauge action of $g \in \Omega^0(U, G)$ is given by

5.2 Optics

Chapter 6

Standard model

Lagrangian field theory

connection as a section?

6.1 (Maxwell equations by action).

$$\mathcal{L} := -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - A_\mu J^\mu.$$

Since

$$\frac{\partial \mathcal{L}}{\partial A_\kappa} = -\frac{\partial(A_\mu J^\mu)}{\partial A_\kappa} = -\frac{\partial A_\mu}{\partial A_\kappa} J^\mu = -\delta_\mu^\kappa J^\mu = -J^\kappa.$$

and

$$\frac{\partial \mathcal{L}}{\partial(\partial_\kappa A_\lambda)} = -\frac{1}{4\mu_0} \frac{\partial(F_{\mu\nu} F^{\mu\nu})}{\partial(\partial_\kappa A_\lambda)} = -\frac{1}{2\mu_0} \frac{\partial(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu)}{\partial(\partial_\kappa A_\lambda)} = -\frac{1}{\mu_0} (\partial^\kappa A^\lambda - \partial^\lambda A^\kappa) = -\frac{1}{\mu_0} F^{\kappa\lambda}$$

because

$$\begin{aligned} F_{\mu\nu} F^{\mu\nu} &= (\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= \partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu - \partial_\nu A_\mu \partial^\mu A^\nu + \partial_\nu A_\mu \partial^\nu A^\mu \\ &= 2(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu) \end{aligned}$$

and

$$\begin{aligned} \frac{\partial(\partial_\mu A_\nu \partial^\mu A^\nu)}{\partial(\partial_\kappa A_\lambda)} &= \frac{\partial(\partial_\mu A_\nu)}{\partial(\partial_\kappa A_\lambda)} \partial^\mu A^\nu + \eta_\mu^\rho \eta_\nu^\sigma \partial_\mu A_\nu \frac{\partial(\partial_\rho A_\sigma)}{\partial(\partial_\kappa A_\lambda)} \\ &= \delta_\mu^\kappa \delta_\nu^\lambda \partial^\mu A^\nu + \eta_\mu^\rho \eta_\nu^\sigma \partial_\mu A_\nu \delta_\rho^\kappa \delta_\sigma^\lambda = 2\partial^\kappa A^\lambda \end{aligned}$$

similarly with

$$\frac{\partial(\partial_\mu A_\nu \partial^\nu A^\mu)}{\partial(\partial_\kappa A_\lambda)} = 2\partial^\lambda A^\kappa,$$

the Euler-Lagrange equation is given by

$$0 = \frac{\partial \mathcal{L}}{\partial A_\kappa} - \partial_\kappa \frac{\partial \mathcal{L}}{\partial(\partial_\kappa A_\lambda)} = -J^\kappa + \frac{1}{\mu_0} \partial_\kappa F^{\kappa\lambda}.$$

6.2 (Noether theorem for classical fields).

$$\mathcal{L} := -\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi$$

Under the translation symmetry $\phi(x) \rightarrow \phi'(x) = \phi(x - \varepsilon)$ with infinitesimal transform parameter vector ε , since we have $\delta\phi = -\varepsilon^\mu \partial_\mu \phi$ from

$$\phi' = \phi - \varepsilon^\mu \partial_\mu \phi + O(\varepsilon^2)$$

and $\delta\mathcal{L} =$ from

$$\mathcal{L}' = \mathcal{L},$$

we can write

$$\begin{aligned} \delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial\phi} \delta\phi + \frac{\partial\mathcal{L}}{\partial\phi^*} \delta\phi^* + \frac{\partial\mathcal{L}}{\partial(\partial\phi)} \delta\partial\phi + \frac{\partial\mathcal{L}}{\partial(\partial\phi^*)} \delta\partial\phi^* \\ &= \end{aligned}$$